

# Genomic Data Security

Yongjin Park

19 March 2026

## OPINIONS ON INTERNET PRIVACY

### THE PHILOSOPHER:

"PRIVACY" IS AN IMPRACTICAL WAY TO THINK ABOUT DATA IN A DIGITAL WORLD SO UNLIKE THE ONE IN WHICH OUR SOCI-

SO BORED.



### THE CRYPTO NUT:

MY DATA IS SAFE BEHIND SIX LAYERS OF SYMMETRIC AND PUBLIC-KEY ALGORITHMS.

WHAT DATA IS IT?

MOSTLY ME EMAILING WITH PEOPLE ABOUT CRYPTOGRAPHY.



### THE CONSPIRACIST:

THESE LEAKS ARE JUST THE TIP OF THE ICEBERG. THERE'S A WAREHOUSE IN UTAH WHERE THE NSA HAS THE ENTIRE ICEBERG.

I DON'T KNOW HOW THEY GOT IT THERE.



### THE NIHILIST:

JOKES ON THEM, GATHERING ALL THIS DATA ON ME AS IF ANYTHING I DO MEANS ANYTHING.



### THE EXHIBITIONIST:

MMMM? I SURE HOPE THE NSA ISN'T WATCHING ME BITE INTO THESE JUICY STRAWBERRIES!!

OOPS I DRIPPED SOME ON MY SHIRT! BETTER TAKE IT OFF.

GOOGLE, ARE YOU THERE?

GOOGLE, THIS LOTION FEELS SOOOO GOOD.



### THE SAGE:

I DON'T KNOW OR CARE WHAT DATA *ANYONE* HAS ABOUT ME.

DATA IS IMAGINARY. THIS BURRITO IS REAL.



# Today's lecture

---

- 1 Genetic Data Utility vs. Privacy
- 2 Membership inference
- 3 Re-identification attack
- 4 Secured protocols for genomic data analysis
- 5 Differential Privacy
- 6 Summary

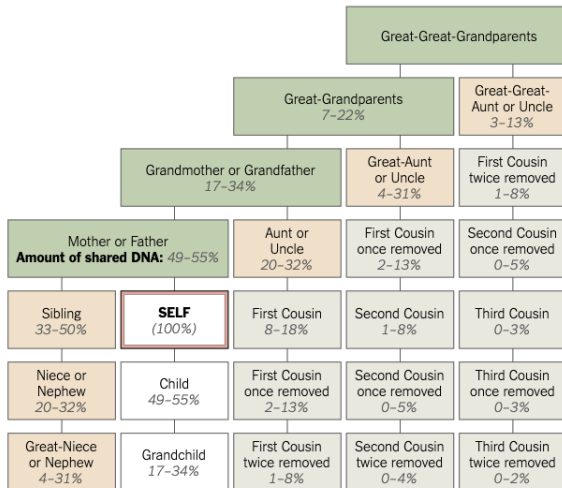


## How Genetic Sleuthing Helped a Kidnapped Girl Recover Her Identity

With the help of a genomic database. . .

<https://www.nytimes.com/2018/10/15/science/gedmatch-genealogy-cold-cases.html>

# Math based on (random) DNA recombination



Source: Blaine T. Bettinger

<https://www.nytimes.com/2018/10/15/science/gedmatch-genealogy-cold-cases.html>

# The Golden State Killer

---

- **Joseph James DeAngelo** arrested in 2018 after 40+ years
- At least 13 murders, 50+ rapes across California (1970s-80s)
- First major cold case solved using Investigative Genetic Genealogy

# Back story: How the case was solved



Applications

Education ▼

DNA Diaries

About ▼

## Where DNA Connections Come Together

**GEDmatch provides more applications for comparing your DNA test results with the most people worldwide, for free!**

[VIEW OUR APPLICATIONS](#)



# Back story: How the case was solved

---

- ① Crime scene DNA uploaded to **GEDmatch.com**



# Back story: How the case was solved

---

- ① Crime scene DNA uploaded to **GEDmatch.com**
- ② Found matches to **distant relatives** (not the suspect)

# Back story: How the case was solved

---

- ① Crime scene DNA uploaded to **GEDmatch.com**
- ② Found matches to **distant relatives** (not the suspect)
- ③ Reconstructed family trees back 5-8 generations

# Back story: How the case was solved

---

- ① Crime scene DNA uploaded to **GEDmatch.com**
- ② Found matches to **distant relatives** (not the suspect)
- ③ Reconstructed family trees back 5-8 generations
- ④ Narrowed down the pool of suspects to a handful of living individuals
- ⑤ Confirmed by collecting discarded DNA (coffee cup) from DeAngelo

# How an Unlikely Family History Website Transformed Cold Case Investigations

Fifteen murder and sexual assault cases have been solved since April with a single genealogy website. This is how GEDmatch went from a casual side project to a revolutionary tool.



# Sooner or Later Your Cousin's DNA Is Going to Solve a Murder

The Golden State Killer case was just the start. Hundreds of cold cases are hot again thanks to a new genealogy technique. The price may be everyone's genetic privacy.

# A tale from the other side of the GSK case

---

## Routes for breaching and protecting genetic privacy

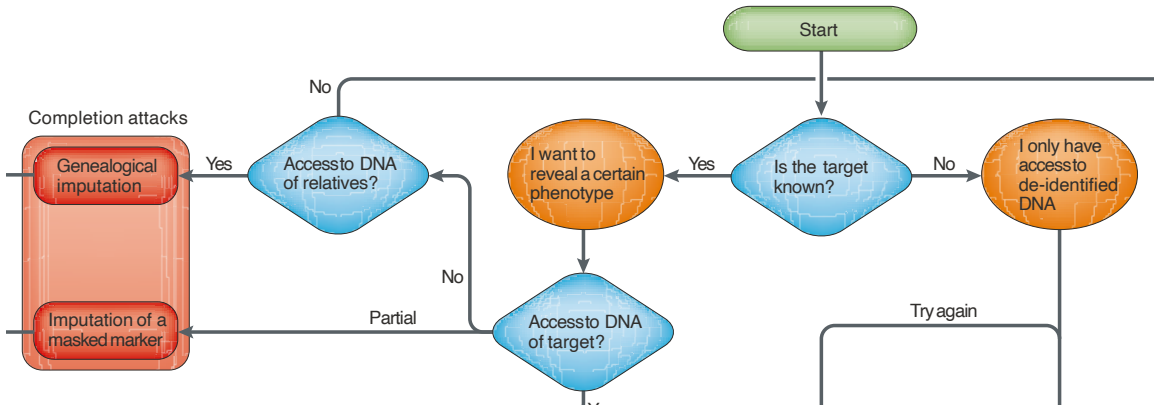
---

*Yaniv Erlich<sup>1</sup> and Arvind Narayanan<sup>2</sup>*

- 60% success with database covering just 2% of population
- As databases grow, nearly anyone can be identified
- Privacy erosion inevitable?

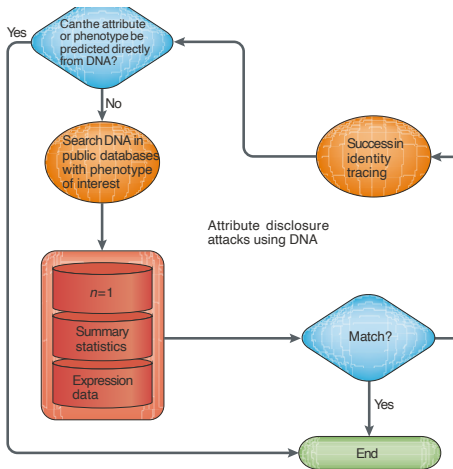
Erlich et al. (2018)

# Key security breach is to find someone in DB



Erlich et al. (2018)

# Genome-Wide Association Studies data could...

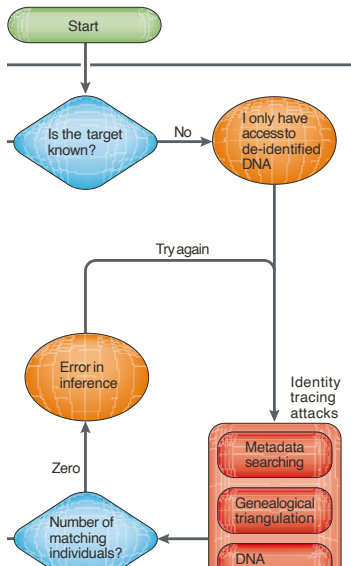


- GWAS summary statistics
- What if we can “look up” (or predict) phenotypes?
- Public genealogical records amplify genetic databases.

Erlich et al. (2018)



# Narrowing down: from genealogy to individual



Yongjin Park

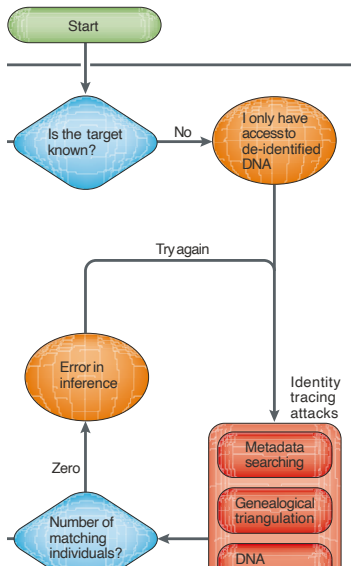
- Demographic identifiers narrow search: genealogy match → age/location → individual
- We can also narrow down to the most genomically informative locations. . .

[https://en.wikipedia.org/wiki/Identity\\_by\\_descent](https://en.wikipedia.org/wiki/Identity_by_descent)

Genomic Data Security

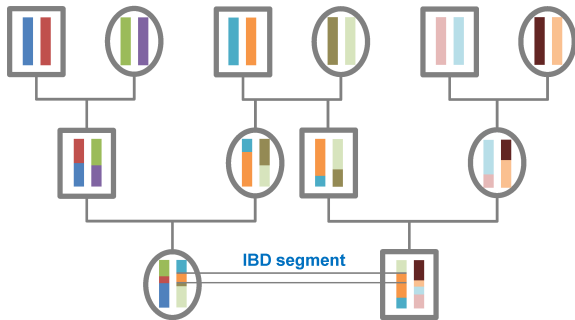
19 March 2026

# Narrowing down: from genealogy to individual



Yongjin Park

- Demographic identifiers narrow search: genealogy match  $\rightarrow$  age/location  $\rightarrow$  individual
- We can also narrow down to the most genomically informative locations...

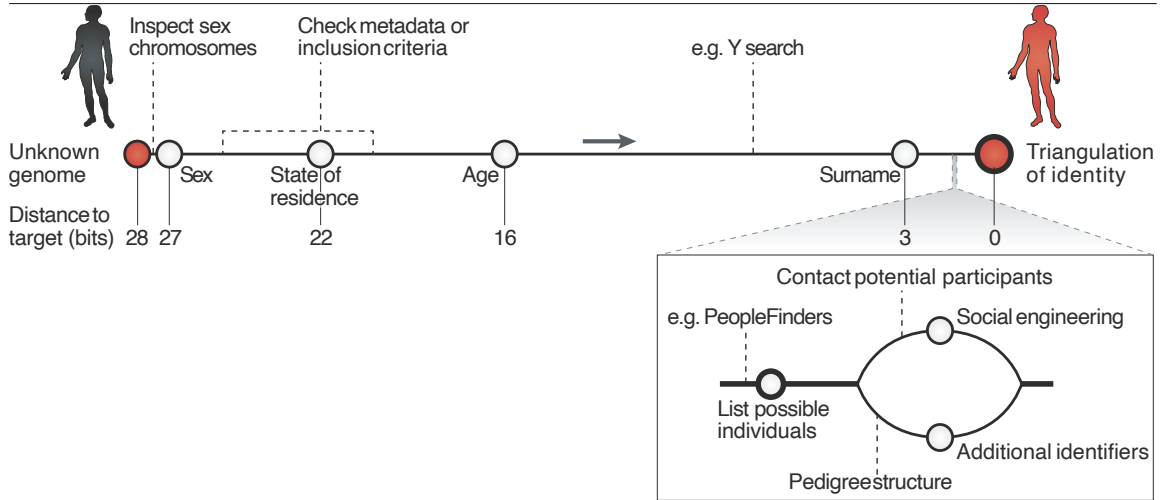


[https://en.wikipedia.org/wiki/Identity\\_by\\_descent](https://en.wikipedia.org/wiki/Identity_by_descent)

Genomic Data Security

19 March 2026

# Real Cases: From DNA to Identification



Erlich et al. (2018)

## ***23andMe Files for Bankruptcy Amid Concerns About Security of Customers' Genetic Data***

The company filed for Chapter 11 bankruptcy to “facilitate a sale process.” Its chief executive stepped down and said she would bid on it.

## ***23andMe to Be Bought by Biotech Company for \$256 Million***

Regeneron Pharmaceuticals said it would acquire 23andMe, which filed for bankruptcy in March and would continue to offer consumer DNA testing services.

- 15 million customers' genetic data  $\approx$  \$ 256M?

## ***23andMe Customers Did Not Expect Their DNA Data Would Be Sold, Lawsuit Claims***

The genetic-testing company, which collected DNA data from users, is for sale in bankruptcy court. Now, 27 states and the District of Columbia oppose selling the data without express consent.

# Privacy Concerns in the midst of corporate dynamics

---

- Genetic data could be sold as an asset to pay creditors
- Should genetic information have special protections in bankruptcy?
- Commercial partnerships with pharma companies for drug discovery
- Who owns data?

# ***REGENERON***®

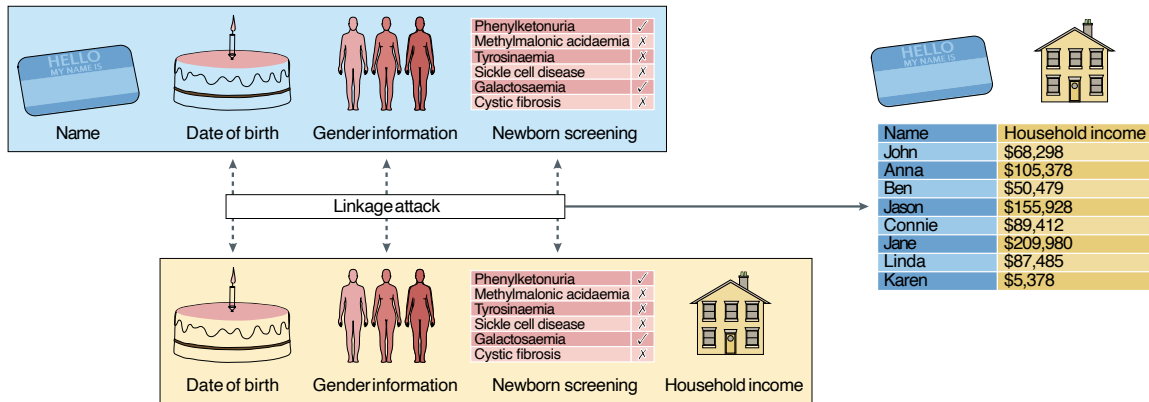
## **Regeneron Enters into Asset Purchase Agreement to Acquire 23andMe® for \$256 Million; Plans to Maintain Consumer Genetics Business and Advance Shared Goals of Improving Human Health and Wellness**

May 19, 2025 at 7:07 AM EDT

**Update:** On July 14, 2025, TTAM Research Institute completed the acquisition of 23andMe assets through a court-supervised bankruptcy process. Regeneron was originally declared the successful bidder in the bankruptcy auction, but once bidding was reopened, declined to submit a higher offer based on its assessment of 23andMe's remaining value.

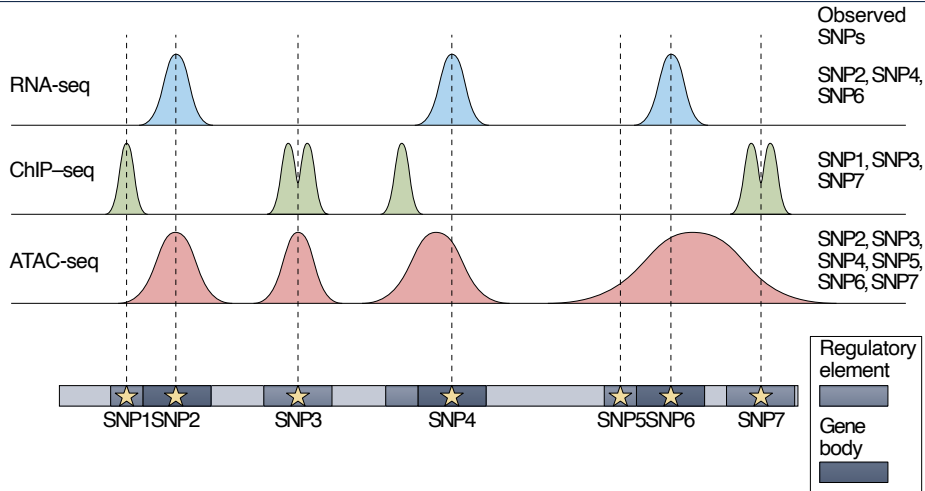


# How genetic data can compromise privacy



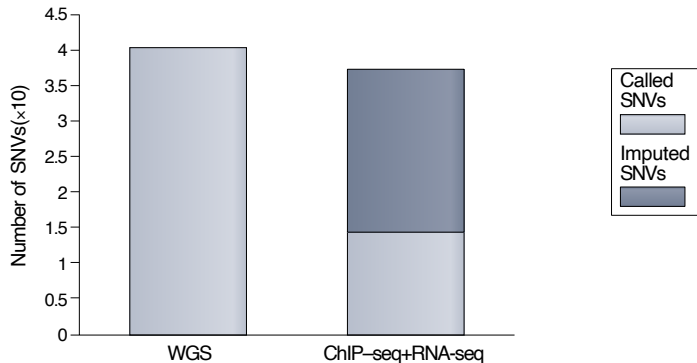
Gürsoy et al. (2022)

# We may be able to infer genotypes from . . .



Gürsoy et al. (2022)

# We could gain more information than we intended



Gürsoy et al. (2022)

- Single Nucleotide Variants
- Although we didn't want to know SNVs, we would be able to infer them.
- Almost every genomic technology involves "sequencing."

# Knowing what the genotypes are ... could open an “unprecedented” possibility?

G → ?

| GWASvariant ID | GWAScharacterizing phenotype | Hi-C |               | Total RNA-seq |               | poly(A) RNA-seq |               | All ChIP-seq |               |
|----------------|------------------------------|------|---------------|---------------|---------------|-----------------|---------------|--------------|---------------|
|                |                              | Raw  | w/ imputation | Raw           | w/ imputation | Raw             | w/ imputation | Raw          | w/ imputation |
| rs1667394      | Blue vsbrown eye             | Yes  | Yes           | Yes           | Yes           | No              | No            | No           | Yes           |
| rs12896399     | Brown vsblond hair           | Yes  | Yes           | No            | Yes           | No              | No            | Yes          | Yes           |
| rs1805007      | Red hair                     | Yes  | Yes           | Yes           | Yes           | No              | No            | No           | No            |
| rs11648785     | Freckles                     | No   | Yes           | Yes           | Yes           | No              | Yes           | No           | Yes           |

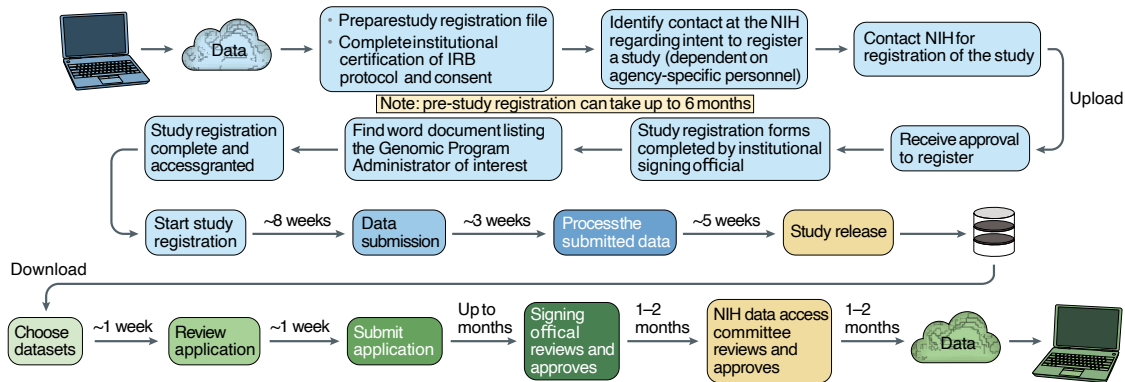
Can variant be genotyped?

Yes

No

Gürsoy et al. (2022)

# After all this... are we safe?



Gürsoy et al. (2022)

# Many outstanding questions...

---

- Should we write “no technology” manifesto like “Unabomber?”
  - What if an insurance company attempts to “predict?”
  - What if we can link them to social media?
- 
- ① How genomic data security can be breached
  - ② How to prevent unwanted information leakage

# Today's lecture

---

- 1 Genetic Data Utility vs. Privacy
- 2 Membership inference**
- 3 Re-identification attack
- 4 Secured protocols for genomic data analysis
- 5 Differential Privacy
- 6 Summary

# Membership attack in genomic data

| Snp | Allele Frequency ( $Y_{i,j}$ )                                                    |                                                                                   |                                                                                   |                                                                                   |                                                                                   | Distance Measure<br>$D(Y_{i,j}) =  Y_{i,j} - \text{Pop}_j  -  Y_{i,j} - M_j $ | Interpretation at the given SNP                                     |
|-----|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------|
|     | 0.0                                                                               | 0.25                                                                              | 0.50                                                                              | 0.75                                                                              | 1.0                                                                               |                                                                               |                                                                     |
| j   |                                                                                   |  |                                                                                   |  |  | $=  1.0 - 0.25  -  1.0 - 0.75 $<br>$= 0.75 - 0.25$<br>$= 0.50$                | most likely to be in the Mixture                                    |
| j+1 |                                                                                   |  |  |  |                                                                                   | $=  0.50 - 0.250  -  0.50 - 0.75 $<br>$= 0.25 - 0.25$<br>$= 0.00$             | equally likely to be in the Mixture and in the Reference Population |
| j+2 |  |  |                                                                                   |  |                                                                                   | $=  0.00 - 0.25  -  0.00 - 0.75 $<br>$= 0.25 - 0.75$<br>$= -0.50$             | most likely to be in the Reference Population                       |

Person of Interest ( $Y_i$ )

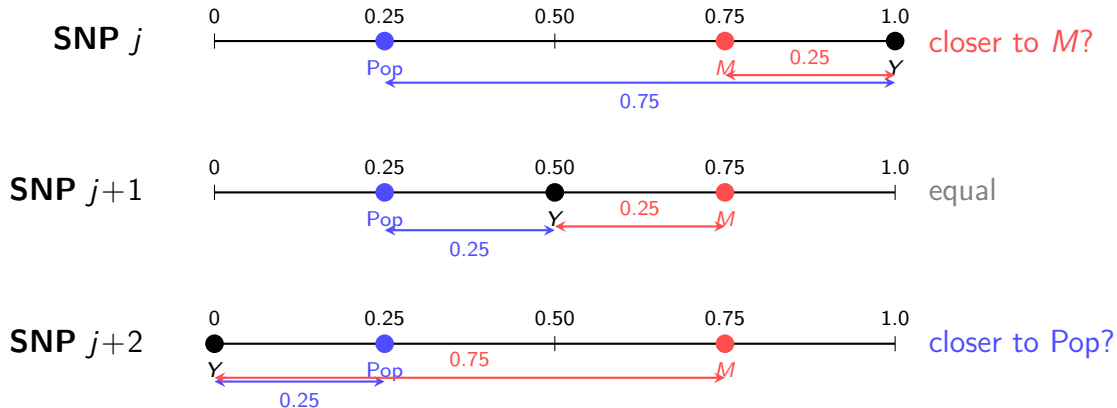
Reference Population (Pop)

Mixture (M)

Homer et al. (2008)



# Basic idea 1: Membership inference

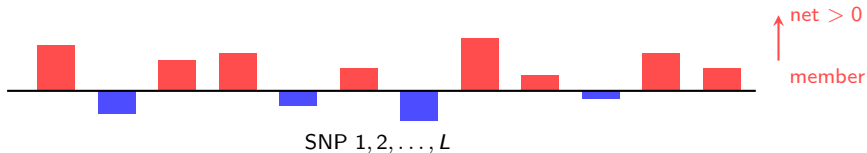


**Q: Is the person of interest in the mixture pool?**

## Basic idea 2: Membership inference

Per SNP:  $D_j = |Y_j - \text{Pop}_j| - |Y_j - M_j|$

Sum:



Test statistic:

$$T = \sum_{j=1}^L D_j \quad \begin{cases} \approx 0 & \text{non-member} \\ > 0 & \text{member} \end{cases}$$

# The test statistic

---

Deviation of target from population at SNP  $j$ :

$$d_j = x_j - 2p_j$$

Inner product with the mixture's deviation:

$$D_{\text{mix}} = \sum_{j=1}^L \underbrace{(f_j^{\text{mix}} - p_j)}_{\text{mixture shift}} \cdot \underbrace{d_j}_{\text{target deviation}}$$

**Decision rule:** declare “member” if  $D_{\text{mix}} > D_{\text{ref}}$

# Why it works

---

$H_0$  (**not in pool**):  $f_j^{\text{mix}}$  and  $x_j$  are independent

$$E[D_{\text{mix}}] \approx 0$$

$H_1$  (**in pool of  $n$** ): target pulls  $f_j^{\text{mix}}$  toward  $x_j$

$$f_j^{\text{mix}} = \frac{1}{2n} \left( x_j + \sum_{i \neq \text{target}} x_{ij} \right)$$

$$E[D_{\text{mix}} \mid H_1] \approx \frac{1}{2n} \sum_{j=1}^L d_j^2 > 0$$

# Membership Test in Code

---

```
## Homer's membership inference test
## Target genotype: x (0, 1, or 2 for each SNP)

## Mixture allele freq: f_mix
## Population allele freq: p_pop

## Target's deviation from population
d <- x - 2 * p_pop

## Test statistic: inner product
D_mix <- sum((f_mix - p_pop) * d)

## Decision
is_member <- D_mix > threshold # TRUE if in pool
```

**Key insight:** Aggregated statistics leak membership

# Signal-to-Noise Ratio

---

$$\text{SNR} \sim \frac{\sqrt{L}}{n}$$

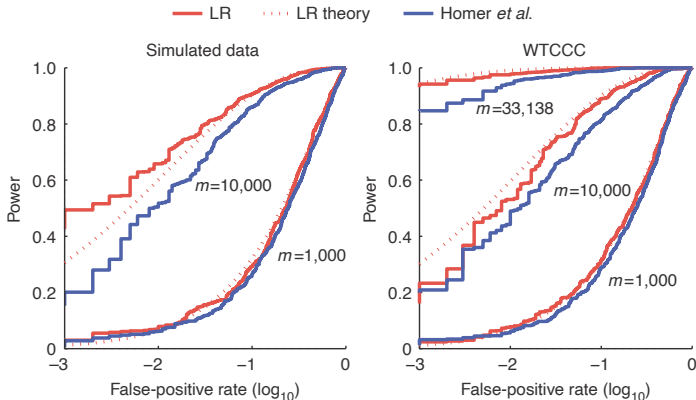
| $L$ (SNPs) | $n$ (pool) | SNR         | Detectable? |
|------------|------------|-------------|-------------|
| 500K       | 100        | $\sim 7$    | Yes         |
| 500K       | 1,000      | $\sim 0.7$  | Marginal    |
| 500K       | 10,000     | $\sim 0.07$ | No          |

With  $L = 500\text{K}$  common SNPs, membership detectable in pools up to  $n \approx 1000$  (Sankararaman et al. 2009)

# Fairly strong membership prediction

## Genomic privacy and limits of individual detection in a pool

Sriram Sankararaman<sup>1,5</sup>, Guillaume Obozinski<sup>2,5</sup>,  
Michael I Jordan<sup>1,2</sup> & Eran Halperin<sup>3,4</sup>



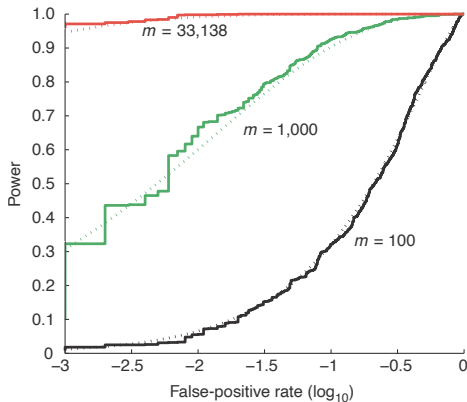
- Pool size  $n = 1,000$
- LR outperforms Homer et al.
- More SNPs ( $m$ )  $\Rightarrow$  higher power
- Theory  $\approx$  simulation
- Validated on real WTCCC data

# Fairly strong membership prediction

Result is **not population-specific** (HapMap YRI):

- Same pattern in African population
- $m = 100$ : near-random
- $m = 1,000$ : moderate power
- $m = 33,138$ : near-perfect

Theory matches empirical across populations



HapMap YRI (African) dataset



# The Beacon Attack

A **beacon** answers one query:

*“Does anyone carry allele A at position X?”*

Response: **yes** / **no** (1 bit)

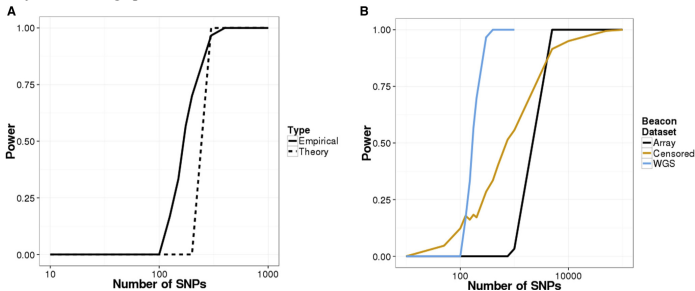
Even this minimal interface leaks membership via repeated queries  
+ LR test

Shringarpure and Bustamante (2015)

ARTICLE

## Privacy Risks from Genomic Data-Sharing Beacons

Suyash S. Shringarpure<sup>1,\*</sup> and Carlos D. Bustamante<sup>1,\*</sup>



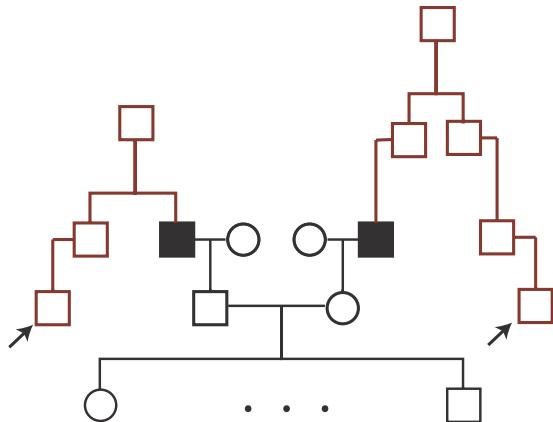
# Identifying Personal Genomes by Surname Inference

Melissa Gymrek,<sup>1,2,3,4</sup> Amy L. McGuire,<sup>5</sup> David Golan,<sup>6</sup> Eran Halperin,<sup>7,8,9</sup> Yaniv Erlich<sup>1\*</sup>

- What's so special about Y chromosome?
- Y-STRs co-segregate with surnames; ~12% of US Caucasian males can be identified
- How does surname frequency affect vulnerability?
- Can rare surnames protect privacy?

# Why Y-STR? No recombination in the patrilineal line

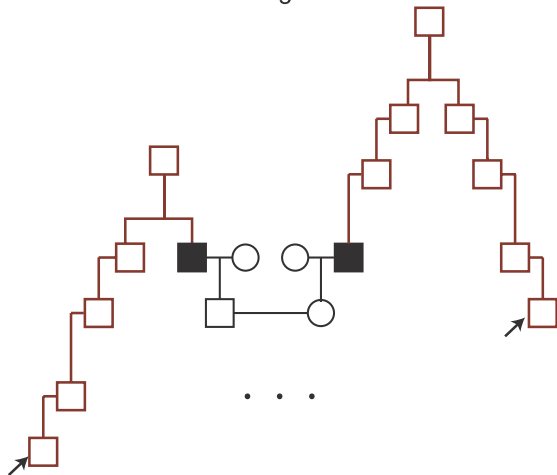
Pedigree 1



- Y chromosome does not recombine  $\Rightarrow$  Y-STR haplotype passed intact father  $\rightarrow$  son
- Red: all patrilineal males share the same Y-STR profile; Black filled: sequenced individuals
- **Common surname** scenario: large lineage, many carriers in genealogy DB
- Easy to find a Y-STR match, but harder to narrow down to one person

# Common surname: Y-STR reaches distant relatives

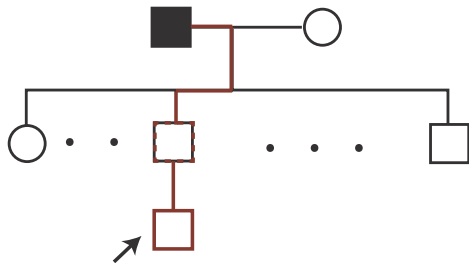
Pedigree 2



- Arrows: individuals matched via genealogy DB, 2-7 meioses from the sequenced person
- Identical Y-STR links distant relatives who never consented to data sharing
- Large pedigree  $\Rightarrow$  many males share the surname, requiring additional metadata to identify

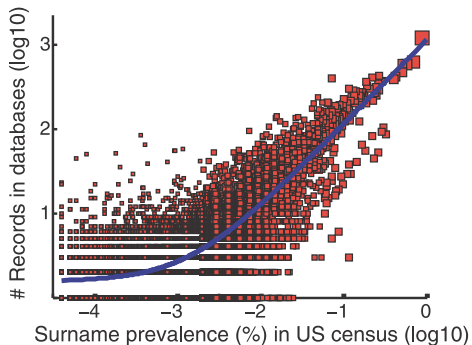
# Rare surname: a single match is nearly unique

Pedigree 3



- Small lineage: few males carry this Y-STR haplotype
- Dashed: genealogy DB record holder; Arrow: identified descendant
- One match in DB  $\Rightarrow$  probability of random match: 1 in  $10^5$  to  $5 \times 10^9$
- Rare surnames are the most vulnerable to re-identification

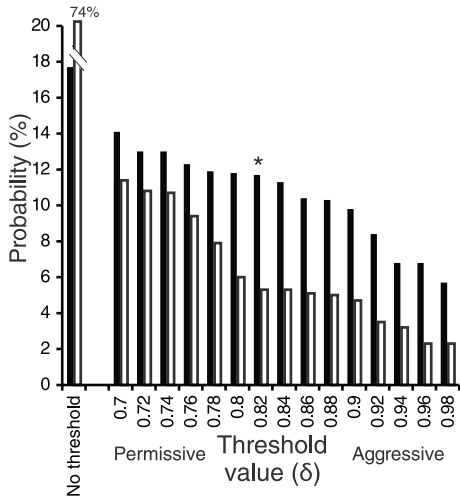
# Surname coverage in genealogy databases



- Genealogy DB coverage tracks US census surname frequency
- Common surnames: many DB records  $\Rightarrow$  easier to find a Y-STR match, but harder to narrow down to one person
- Rare surnames: fewer DB records, but a single match is highly identifying
- The attack surface grows as hobbyist databases expand

Gymrek et al. (2013)

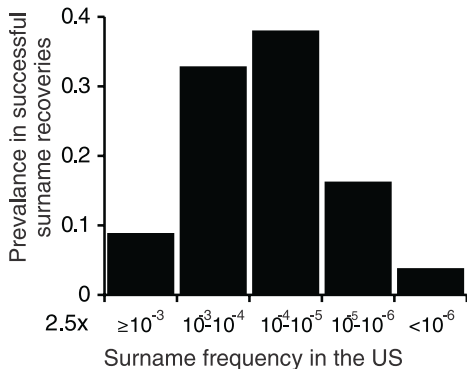
# Demographic Triangulation



- Black: surname recovery rate; Gray: false positive rate
- Trade-off: at  $\delta = 0.82$ , ~12% success with ~5% FPR
- Surname + age + state narrows 60,000 candidates  $\rightarrow$  median of 12 individuals
- How much metadata is too much?
- What should be protected beyond HIPAA?

# Real-World Re-identification: Craig Venter

---

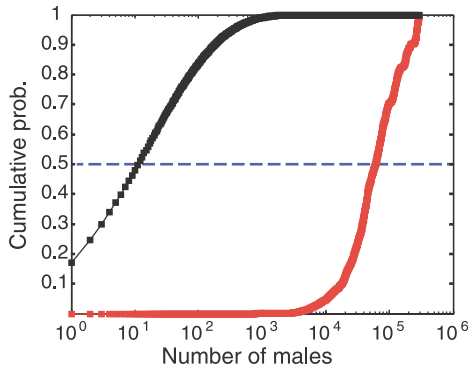


- Wait, where is Craig Venter's DNA data?
- Clear match to "Venter" record in genetic genealogy database
- Public genomes in GenBank can be identified by name

Gymrek et al. (2013)



# How much does Y-STR matching narrow the search?



- Red: candidate pool from census alone (median  $\sim 10^5$  males); Black: after Y-STR surname match (median  $\sim 10$ )
- 99% accuracy from Illumina sequencing; works even at  $10\times$  coverage
- Technical noise alone does not prevent the attack

Gymrek et al. (2013)

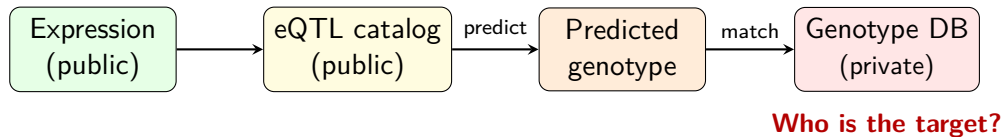
# Today's lecture

---

- 1 Genetic Data Utility vs. Privacy
- 2 Membership inference
- 3 Re-identification attack**
- 4 Secured protocols for genomic data analysis
- 5 Differential Privacy
- 6 Summary

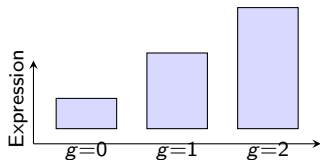
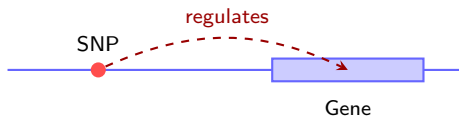
# eQTL Linking Attacks

---



# What is eQTL?

An **expression quantitative trait locus (eQTL)** is a genomic variant (SNP) whose genotype is associated with the expression level of a gene.



$$e = \beta \cdot g + \epsilon$$

- $g \in \{0, 1, 2\}$ : genotype (minor allele count)
- $e$ : expression level
- $\beta$ : eQTL effect size

# Quantification of private information leakage from phenotype-genotype data: linking attacks

Arif Harmanci<sup>1,2</sup> & Mark Gerstein<sup>1-3</sup>

## Input:

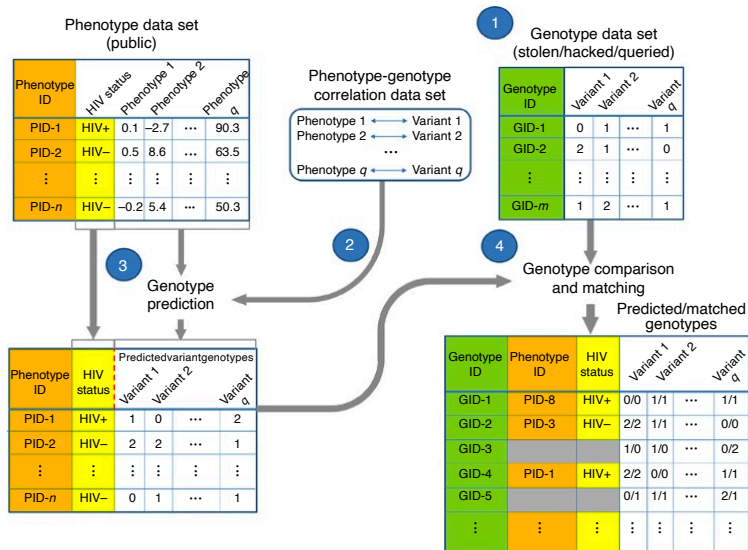
- Expression data
- eQTL summary statistics

## Output:

- Genotypes

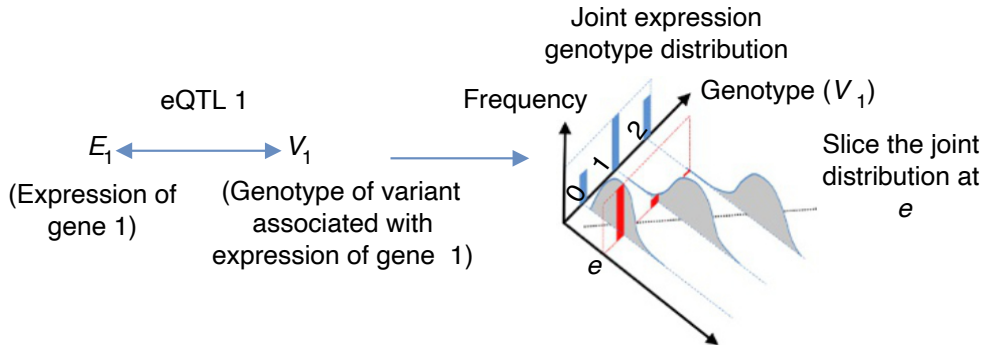
**Implication:**  $G \rightarrow ?$

# Genotypes are predictable from expression



- Correlation info links expression to variants
- Predict genotypes from observed expression
- Genotypes  $\Rightarrow$  re-identification
- Would involuntarily reveal sensitive information!

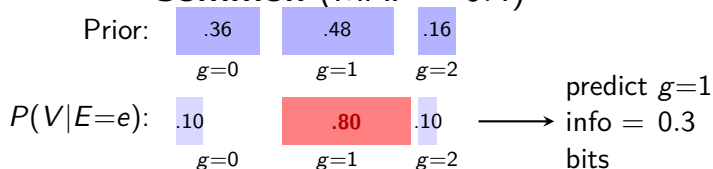
# Using expression QTL as backdoor...



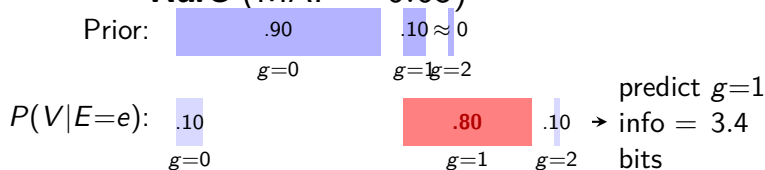
- eQTL defines a joint distribution of expression ( $E$ ) and genotype ( $V$ )
- Given observed expression  $e$ , slice the joint distribution  $\Rightarrow$  conditional  $P(V|E = e)$
- Strong eQTL: the slice is peaked at one genotype  $\Rightarrow$  high information leakage

# Tradeoff: predictability vs. identifiability

## Common (MAF = 0.4)



## Rare (MAF = 0.05)



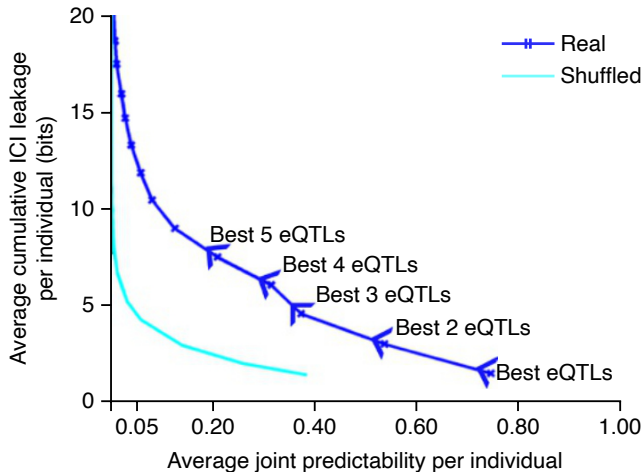
Conditional self-information:

$$\log_2 \frac{1}{P(V = g \mid E = e)}$$

- Both predict  $g=1$  with 80% confidence
- Common:  $g=1$  has prior .48  $\Rightarrow$  low surprise
- Rare:  $g=1$  has prior .10  $\Rightarrow$  high surprise, highly identifying



# Three-Step Linking Framework



- 1 Select eQTLs (correlation threshold)
  - 2 Predict genotypes from expression
  - 3 Link to database via distance metric
- Adding gender, population improves accuracy to  $>95\%$
  - Even genotypes alone achieve  $\sim 85\%$  accuracy

(Harmanci and Gerstein 2016)

### Step 1

G-P  
correlation  
data set

Selection

- Absolute value of correlation

### Step 2

Prediction  
methodology

Genotype  
prediction

- Maximum *a posteriori* genotype

### Step 3

- Gender, population, age

Auxiliary  
information

Linking

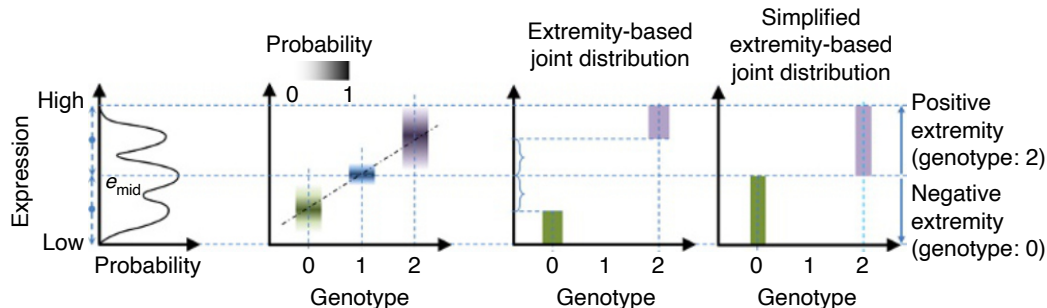
- Minimum distance between predicted and individual genotypes

Estimate  
reliability of  
linking

- First distance gap



# Simplified attack: extremity-based prediction



- Don't need the full joint distribution: just use extreme expression values
- High expression  $\Rightarrow$  predict  $g = 2$ ; Low expression  $\Rightarrow$  predict  $g = 0$
- Robust across eQTL selection criteria; PPV  $> 95\%$  for most thresholds

# A working example on eQTL linkage attack

## Attacker has (public):

- Target's expression:  $\mathbf{e} = (e_1, \dots, e_K)$
- eQTL catalog: Gene  $k \leftrightarrow$  SNP  $k$ ,  
effect  $\beta_k$

## Attacker wants:

- Predicted genotype:  
 $\hat{g}_k = \text{round}(e_k / \beta_k)$
- Match  $\hat{\mathbf{g}}$  to a row in the genotype DB

| Gene | Expression | $\beta$ | Predicted $\hat{g}$ |
|------|------------|---------|---------------------|
| 35   | 0.2        | 1.3     | 0                   |
| 27   | 0.8        | 0.8     | 1                   |
| 5    | 0.8        | 0.8     | 1                   |
| 21   | 0.8        | 0.8     | 1                   |
| 23   | 0.7        | 1.0     | 1                   |

Match this vector against  $n = 500$  individuals

- How many unknowns?
- How many pieces of evidence?

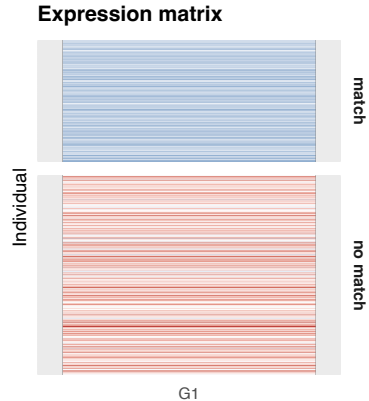
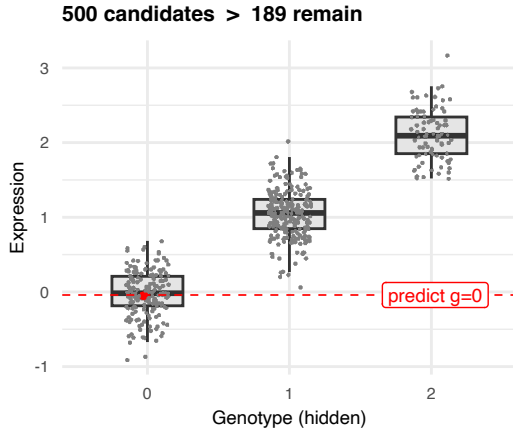
# The Attack in Code

---

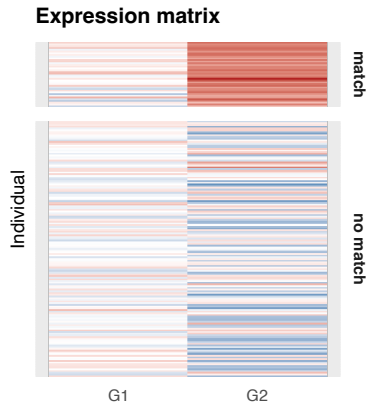
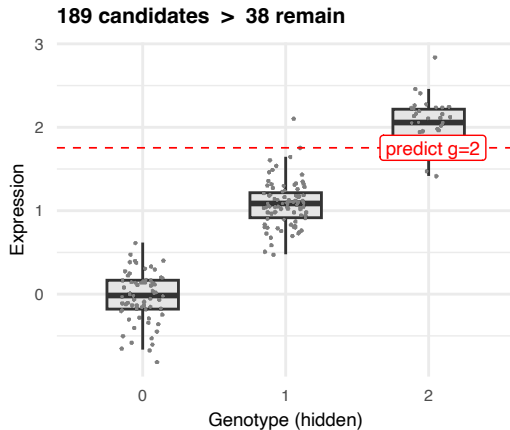
```
## eQTL model: expression = beta * genotype + noise  
E <- beta * G + rnorm(n, sd = sigma)  
  
## Predict genotype from expression (reverse the model)  
pred_genotype <- round(E / beta)  
pred_genotype <- pmin(pmax(pred_genotype, 0), 2) # clip to 0,1,2  
  
## Match predicted genotype to database  
## --> could be just nearest neighbour search  
matches <- which(G_database == pred_genotype)
```

Simple algebra can leak the identity...

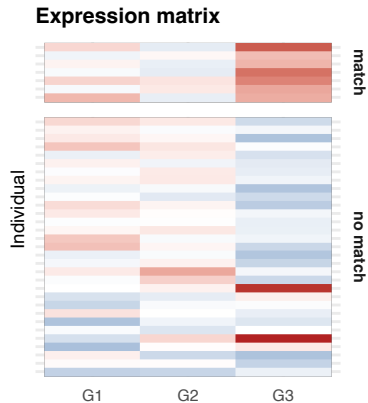
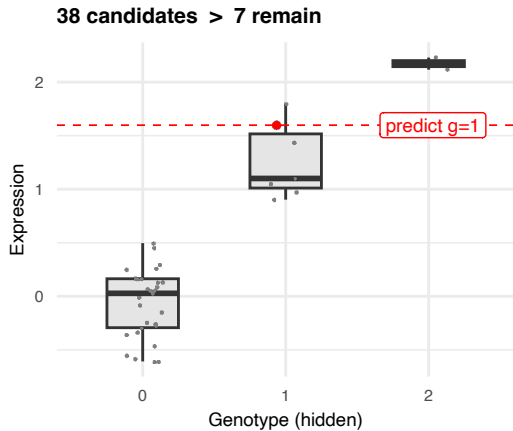
# Gene 1: predicting genotype from expression



# Gene 2: narrowing the pool

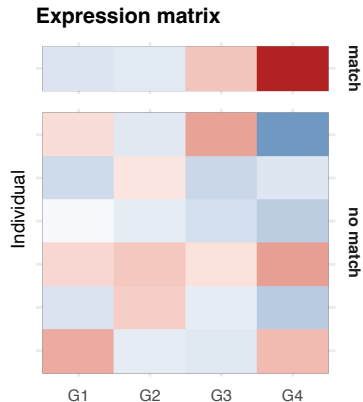
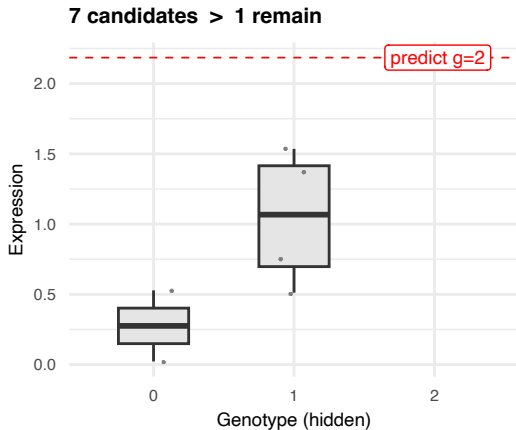


# Gene 3: a rare variant

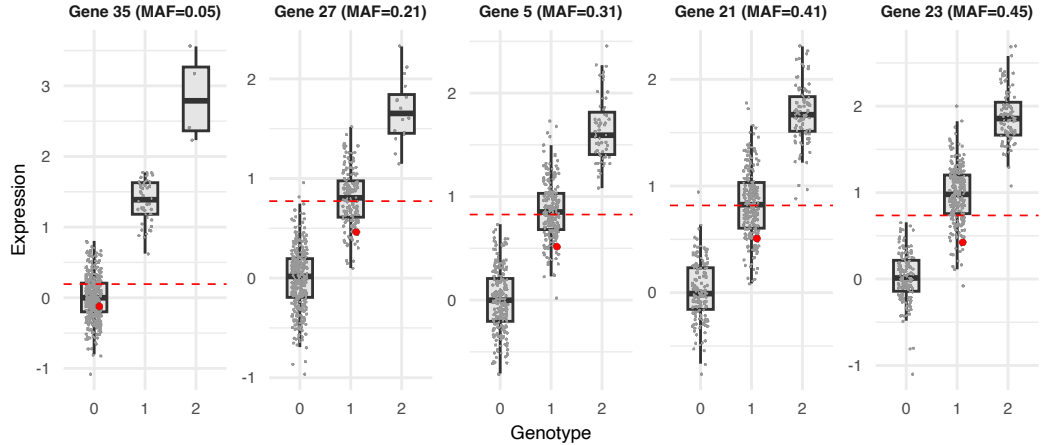




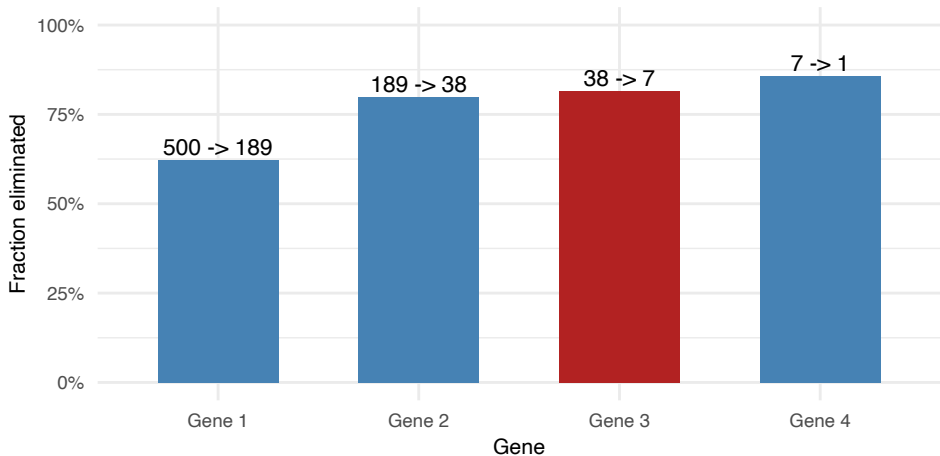
# Gene 4: identified



# Genotype vs. expression: the full picture



# Not all genes are equal



**Q: Why is a rare-variant gene more informative?**

# Weight matches by rarity

---

Exact matching is fragile: one wrong prediction breaks the chain.

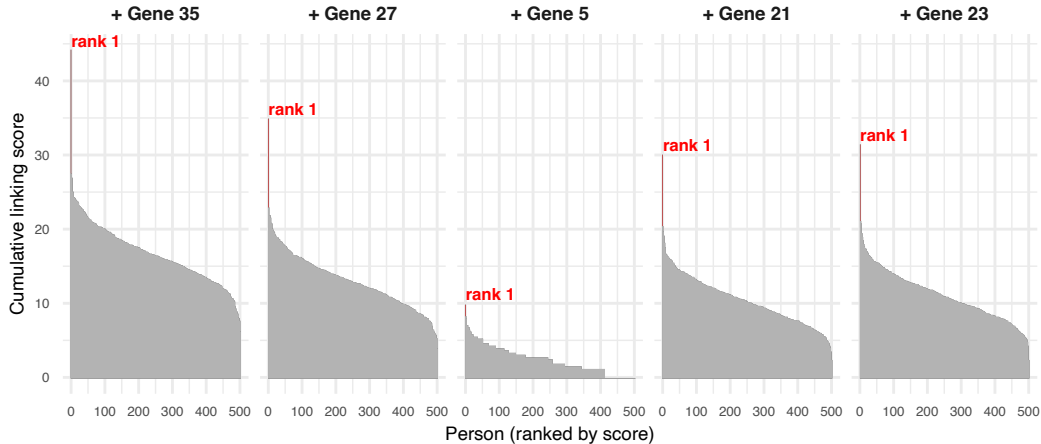
**Better approach:** score each match by how informative it is.

- If 45% of people have genotype  $g = 1$ , matching on it is **weak evidence**
- If 5% of people have genotype  $g = 2$ , matching on it is **strong evidence**

$$S_i = \sum_k \mathbb{I}[G_{ik} = \hat{g}_k] \cdot \underbrace{\log_2(1/f(\hat{g}_k))}_{\text{information content}}$$

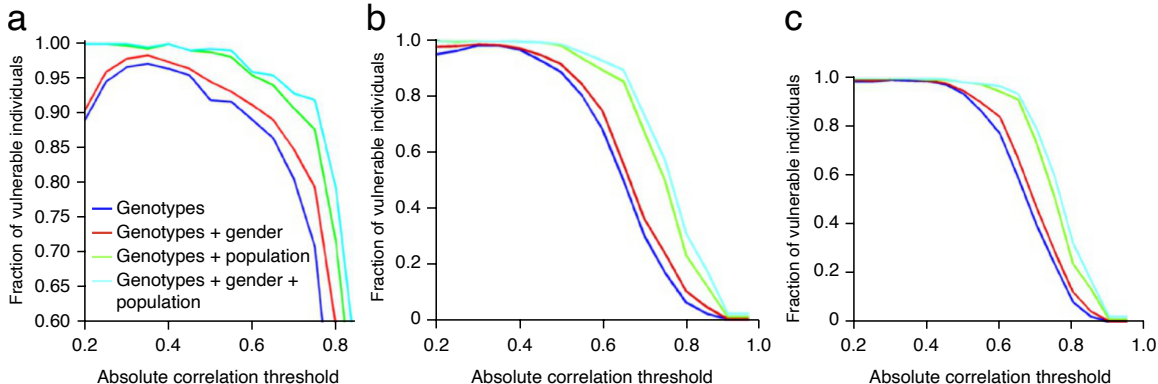
This is just information theory: rare events carry more information.

# Accumulating evidence across genes



Each panel adds one gene. The target (red) moves toward rank 1 as evidence accumulates.

# Auxiliary metadata amplifies the attack

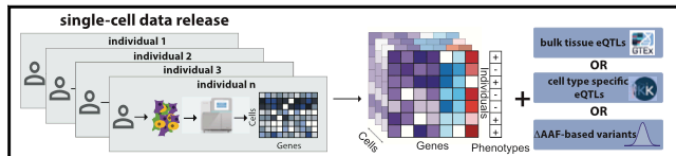


- Genotypes alone: ~85% of individuals vulnerable at moderate eQTL threshold
- Adding gender or population pushes vulnerability above 95%

(Harmanci and Gerstein 2016)

# Private information leakage from single-cell count matrices

## Graphical abstract



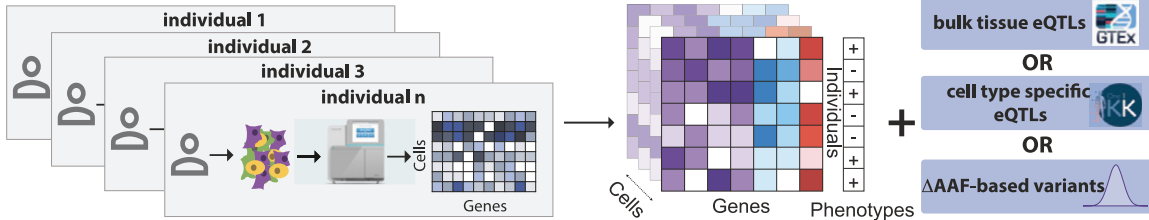
## Authors

Conor R. Walker, Xiaoting Li,  
Manav Chakravarthy,  
William Lounsbery-Scaife,  
Yoolim A. Choi, Ritambhara Singh,  
Gamze Gürsoy

- Linking attack workflow: scRNA-seq → genotype prediction → database matching
- How can we share valuable single-cell data while protecting donor identity?

# Linking with Bulk eQTLs

## single-cell data release



- 93-98% linking accuracy using publicly available tissue eQTLs
- Even “common” cell types leak identity. What counts as rare enough to be safe?

Walker et al. (2024)





### genome database

| IDs                                                                               | genotypes |       |       |  |       |       |
|-----------------------------------------------------------------------------------|-----------|-------|-------|--|-------|-------|
|                                                                                   | $v_1$     | $v_2$ | $v_3$ |  | $v_i$ | $v_m$ |
|  | 2         | 2     | 0     |  | 1     | 0     |
|  | 2         | 2     | 0     |  | 0     | 1     |
|  | 1         | 1     | 2     |  | 0     | 1     |
|                                                                                   |           |       |       |  |       |       |
|  | 2         | 2     | 1     |  | 2     | 2     |
|  | 1         | 0     | 2     |  | 0     | 1     |

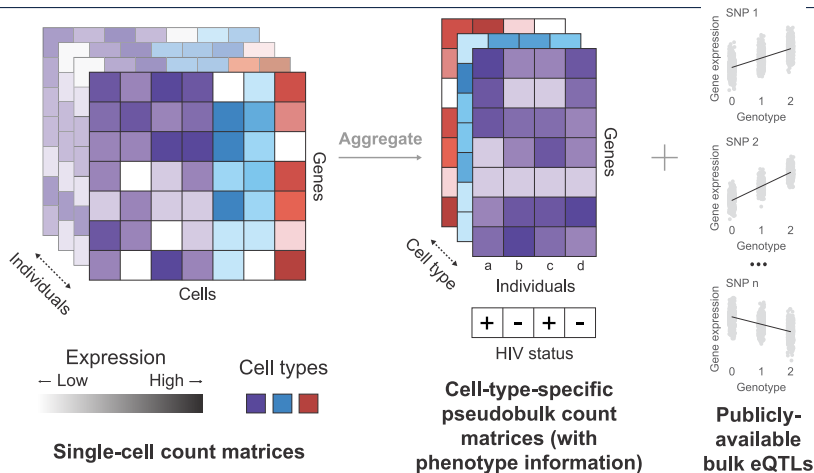
linking



↓  
genotype  
prediction

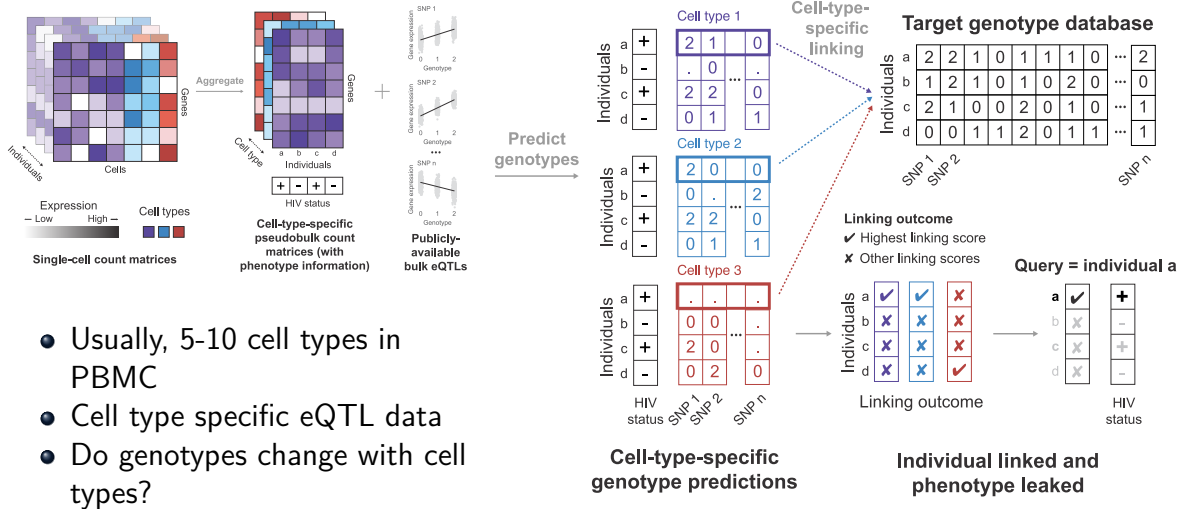
| ind | noisy genotypes |       |       |  |       |       | pheno |
|-----|-----------------|-------|-------|--|-------|-------|-------|
|     | $v_1$           | $v_2$ | $v_3$ |  | $v_i$ | $v_m$ |       |
| 1   | 2               | 2     | 0     |  | ?     | 2     | +     |
| 2   | 2               | 0     | ?     |  | 0     | 0     | -     |
| 3   | 0               | ?     | 2     |  | 0     | ?     | -     |
|     |                 |       |       |  |       |       | +     |
| 10  | 0               | 2     | 2     |  | 2     | 2     | +     |
| n   | 0               | ?     | 2     |  | ?     | ?     |       |

# The way we called eQTLs can create a backdoor



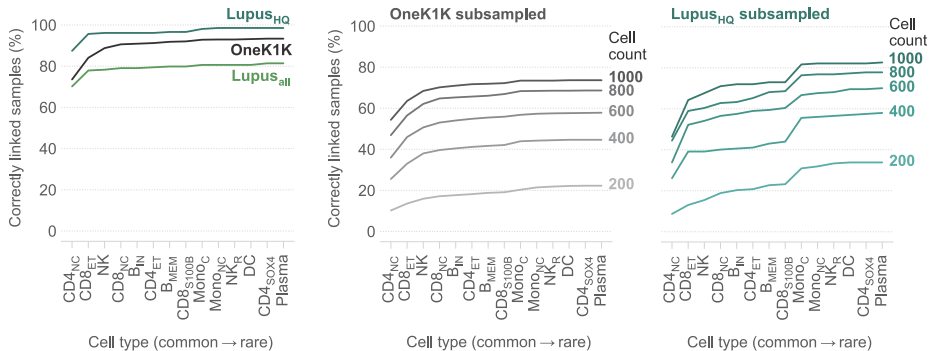
- **Input:** (1) pseudobulk (2) eQTL catalogue

## Well, multiple cell types $\approx$ multiple lines of evidence



Walker et al. (2024)

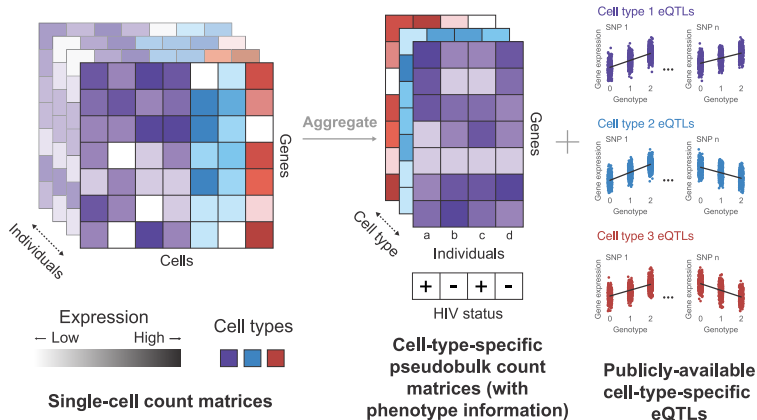
# Revealing “donor” information using external DB!



- Rarer cell types  $\Rightarrow$  higher linking accuracy (lines trend upward, common  $\rightarrow$  rare)
- More cells per type  $\Rightarrow$  better pseudobulk estimates  $\Rightarrow$  better genotype prediction

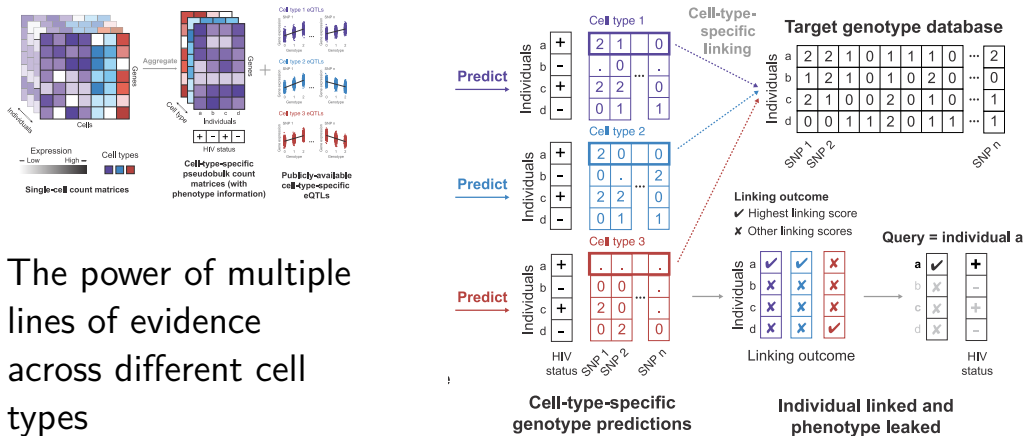
Walker et al. (2024)

# What if we have single-cell data and eQTLs?



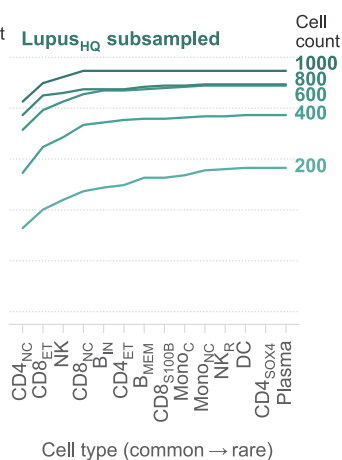
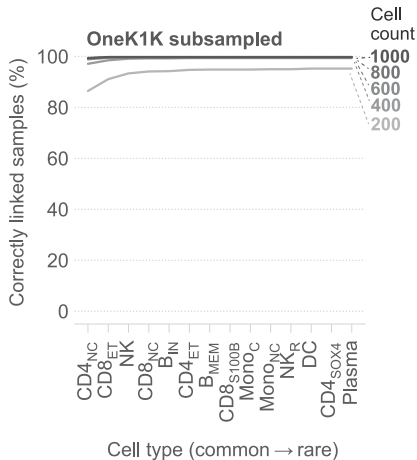
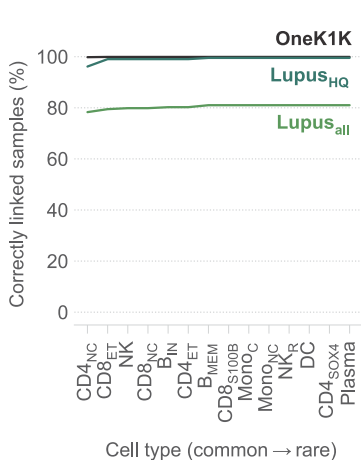
- Can normalization and PCA remove non-genetic individual-specific information?
- Single-cell count matrices aggregated into cell-type-specific pseudobulk

# What if we have single-cell data and eQTLs?



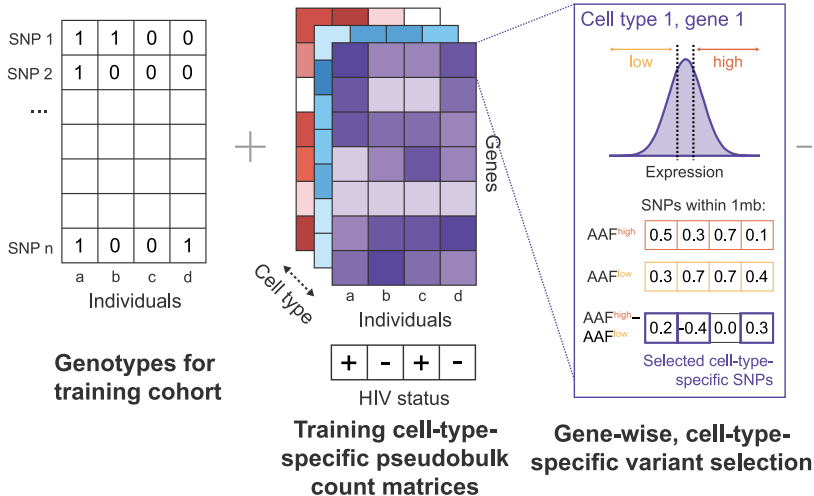
Walker et al. (2024)

# Again, strong prediction performance...



Walker et al. (2024)

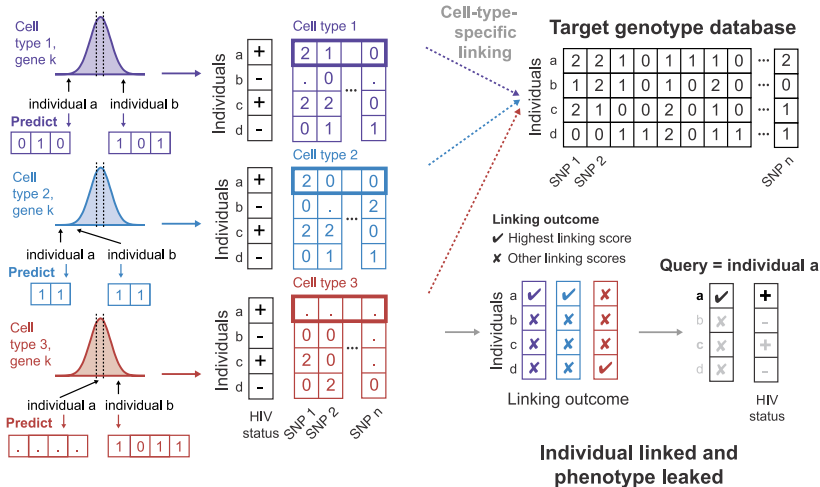
# Do we even need scQTL summary statistics data?



- No external eQTL catalog needed
- AAF = alternative allele frequency
- Split by expression (high/low), compare AAF per locus
- Large  $\Delta\text{AAF} \Rightarrow$  informative variant
- **Cross-cohort:** learn variants from training cohort, attack a different target cohort



# Genotypes in training data → predict other cohorts





# Today's lecture

---

- 1 Genetic Data Utility vs. Privacy
- 2 Membership inference
- 3 Re-identification attack
- 4 Secured protocols for genomic data analysis**
- 5 Differential Privacy
- 6 Summary

---

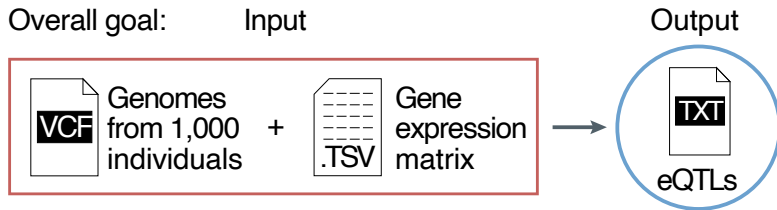
# Functional genomics data: privacy risk assessment and technological mitigation

---

*Gamze Gürsoy , Tianxiao Li , Susanna Liu , Eric Ni ,  
Charlotte M. Brannon  and Mark B. Gerstein *

# How can we safely estimate eQTLs?

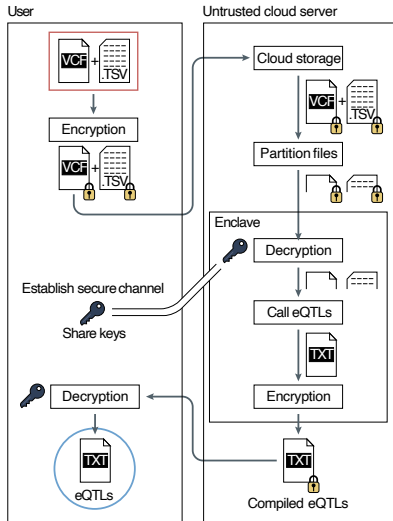
---



**Goal:** Protect expression profiled individuals from re-identification attacks

Gürsoy et al. (2022)

# Trusted Execution Environments (TEE)



- Data encrypted in transit and at rest
- Decrypted only inside hardware **enclave** (e.g., Intel SGX)
- Even the OS cannot read enclave memory

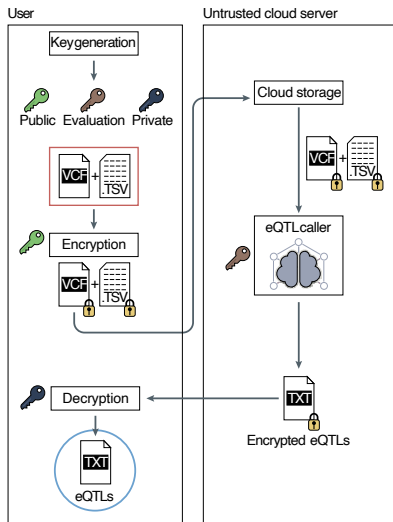
## Pros:

- near-plaintext speed
- no utility loss

## Cons:

- side-channel attacks
- trust in hardware vendor
- limited enclave memory

# Homomorphic Encryption (HE)



Arithmetic on **encrypted data** yields correct results when decrypted:

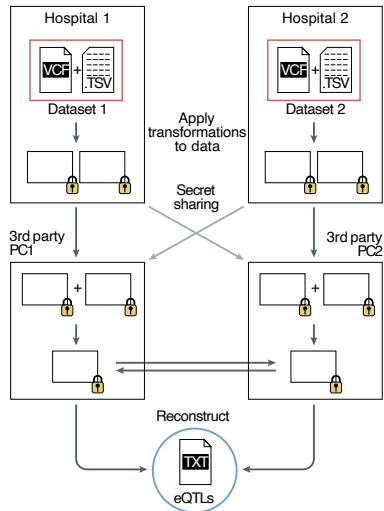
$$\text{Dec}(f(\text{Enc}(x))) = f(x)$$

- Server runs eQTL caller on encrypted genotypes and expression (never sees the actual data)
- Only the user holds the private key to decrypt the result

**Pros:** strong formal guarantee (computational)

**Cons:**  $10^3$ – $10^6 \times$  slowdown vs. plaintext

# Secure Multiparty Computation (MPC)



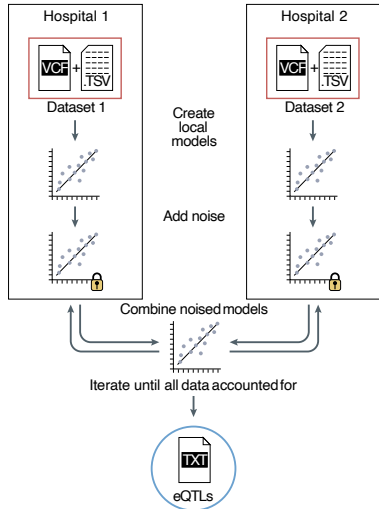
Multiple hospitals pool data **without sharing it**

- Each site splits data into random **secret shares**
- Third-party servers compute on shares
- No single party can reconstruct individual data

**Key property: no utility loss.** Exact same result as pooled plaintext.



# Federated Learning (FL)



Raw data **never leaves** each institution

- 1 Train local models
- 2 Add noise to model updates
- 3 Combine noised models centrally
- 4 Iterate until convergence

**Caveat:** gradients can leak training data. FL alone provides only informal privacy guarantees.

# Today's lecture

---

- 1 Genetic Data Utility vs. Privacy
- 2 Membership inference
- 3 Re-identification attack
- 4 Secured protocols for genomic data analysis
- 5 Differential Privacy**
- 6 Summary

LONG LIVE THE REVOLUTION.  
OUR NEXT MEETING WILL BE  
AT THE DOCKS AT MIDNIGHT  
ON JUNE 28 TAB

AHA, FOUND THEM!



WHEN YOU TRAIN PREDICTIVE MODELS  
ON INPUT FROM YOUR USERS, IT CAN  
LEAK INFORMATION IN UNEXPECTED WAYS.

# Differential Privacy: Formal Privacy Guarantees

---

A randomized mechanism  $M$  satisfies  $\epsilon$ -DP if for all neighboring datasets  $D, D'$ :

$$P(M(D) \in S) \leq e^\epsilon \cdot P(M(D') \in S)$$

**Intuition:** whether or not you participate, published results change by at most  $e^\epsilon$ .

| $\epsilon$ | $e^\epsilon$  | Interpretation      |
|------------|---------------|---------------------|
| 0.1        | 1.11          | Strong privacy      |
| 1          | 2.72          | Moderate            |
| 10         | $\sim 22,000$ | Essentially vacuous |

# DP for GWAS: Adding Noise to Protect Individuals

---

**Private MAF** at a single SNP in  $n$  diploid individuals (Uhler et al. 2012; Yamamoto and Shibuya 2021):

$$\hat{f}_{\text{priv}} = \underbrace{\frac{1}{2n} \sum_{i=1}^n x_i}_f + \text{Lap}\left(\frac{1}{n\varepsilon}\right)$$

- Sensitivity: changing one person shifts  $f$  by at most  $\Delta f = 1/n$
- For  $n = 5,000$ ,  $\varepsilon = 1$ : noise SD  $\approx 0.00028$  vs. sampling error  $\sim 0.01$

**Tradeoff:** DP adds statistical noise. Most painful for **rare variants** where the signal is already small.

# DP in Code: Laplace Mechanism

---

```
## Add Laplace noise for epsilon-differential privacy
## Sensitivity: max change from adding/removing one person

## Allele frequency with DP
n <- 5000
epsilon <- 1.0
f <- mean(genotypes) / 2 # true MAF

## Add Laplace noise: Lap(sensitivity/epsilon)
sensitivity <- 1/n
noise <- rlaplace(1, scale = sensitivity/epsilon)
f_private <- f + noise

## Publish f_private instead of f
```

**Key insight:** Controlled noise gives privacy guarantee

# DP for Association Tests: Protecting $\chi^2$ Statistics

For a GWAS with genotype  $\{0, 1, 2\} \times$  case/control, form a  $3 \times 2$  contingency table (Uhler et al. 2012):

|         | Case     | Control  |
|---------|----------|----------|
| $g = 0$ | $n_{00}$ | $n_{01}$ |
| $g = 1$ | $n_{10}$ | $n_{11}$ |
| $g = 2$ | $n_{20}$ | $n_{21}$ |

**Sensitivity** (Uhler Thm 3.2): Adding or removing one person changes  $\chi^2$  by at most

$$\Delta(\chi^2) = \frac{4N}{N+2}$$

For large  $N$ , this is  $\approx 4$ , while  $\chi^2$  for true associations grows with  $N \Rightarrow$  relative noise shrinks as sample size grows.

# Private $\chi^2$ Release

---

**Private release:**

$$\chi_{\text{priv}}^2 = \chi^2 + \text{Lap}\left(\frac{1}{\varepsilon} \cdot \frac{4N}{N+2}\right)$$

**Example:**  $N = 5,000$ ,  $\varepsilon = 1$

- Sensitivity:  $(4 \times 5000)/5002 \approx 4.0$
- Noise SD:  $4.0/1.0 \times \sqrt{2} \approx 5.7$
- A genome-wide significant SNP has  $\chi^2 \approx 30 \Rightarrow \text{signal-to-noise} \approx 5 : 1$

But do we need to release **all**  $M'$  test statistics? Noise grows with number released.



# Making DP Practical: Use $\log_{10}$ p-Values

---

**Key insight:** release  $-\log_{10}(p)$  instead of  $p$  (Yamamoto and Shibuya 2021).

**Sensitivity** for a  $3 \times 2$  table (Yamamoto Prop. 2):

$$\Delta(-\log_{10} p) = \log_{10}(e) \cdot \frac{2N}{N+2}$$

For  $N = 100,000$ : sensitivity  $\approx 0.87 \Rightarrow$  **manageable noise**

**Private release:**

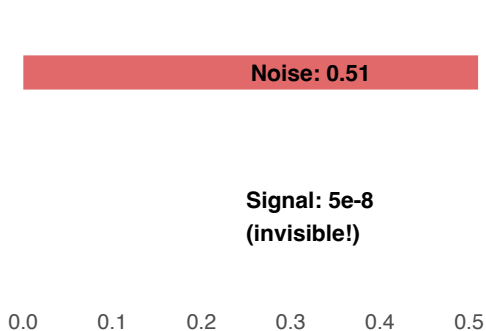
$$(-\log_{10} p)_{\text{priv}} = -\log_{10}(p) + \text{Lap}\left(\frac{0.87}{\epsilon}\right)$$

The significance threshold is now **additive**:  $-\log_{10}(5 \times 10^{-8}) = 7.3$

$\Rightarrow$  Compare noisy  $-\log_{10}(p)$  directly against 7.3

## Raw p-value scale

Signal completely buried by noise



## $-\log_{10}(p)$ scale

Signal clearly exceeds noise!



# Efficient DP: Release Only Top SNPs, Not All

---

**Uhler Algorithm 1** (Uhler et al. 2012) (3 steps,  $\epsilon$ -DP by Thm 3.7):

Let  $S(\chi^2) = 4N/(N+2)$ . Given  $M'$  total SNPs, release top  $M$ :

- ➊ **Noisy ranking:** Add  $\text{Lap}\left(\frac{4M}{\epsilon} \cdot S(\chi^2)\right)$  to all  $M'$  statistics
- ➋ **Select top  $M$ :** Pick the  $M$  SNPs with largest noisy  $\chi^2 \Rightarrow$  set  $\mathcal{S}$
- ➌ **Re-noise the winners:** Add *fresh*  $\text{Lap}\left(\frac{2M}{\epsilon} \cdot S(\chi^2)\right)$  to the **true**  $\chi^2$  values of SNPs in  $\mathcal{S}$

**Key property:** noise scales with  $M$  (number released), **not**  $M'$  (total SNPs tested).

# Safe Z-Score Release via Post-Processing

---

**Post-processing theorem:** Any function of a DP output is also DP - no additional privacy cost.

Since  $Z^2 = \chi^2$  for 1-df tests, derive a private Z-score from the private  $\chi^2$ :

$$Z_{\text{priv}} = \text{sign}(\hat{\beta}) \cdot \sqrt{\max(0, \chi^2_{\text{priv}})}$$

Alternatively, derive from the private  $-\log_{10}(p)$ :

$$Z_{\text{priv}} = \Phi^{-1}\left(\frac{1}{2} \cdot 10^{-(-\log_{10} p)_{\text{priv}}}\right)$$

This lets downstream consumers use **familiar Z-scores** for meta-analysis, fine-mapping, etc. All for free under DP composition.

# Code: Uhler Algorithm 1 + Z-Score Derivation

```
## Uhler Algorithm 1: Release top M SNPs privately
N <- 5000; M <- 100; M_prime <- 500000; eps <- 1.0

## Sensitivity of chi-squared for a 3x2 table
sens_chi2 <- 4 * N / (N + 2)

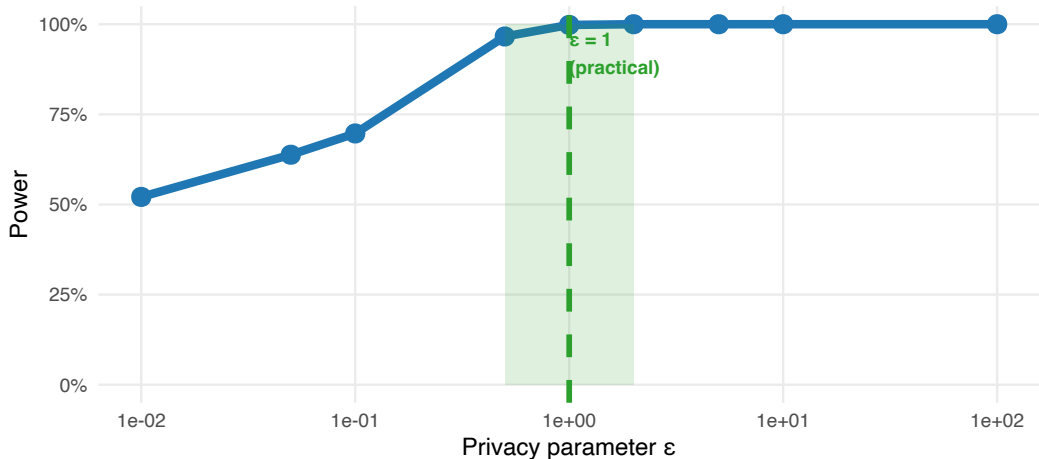
## Step 1: Noisy ranking
chi2_noisy <- chi2_all +
  rlaplace(M_prime, scale = 4 * M / eps * sens_chi2)

## Step 2: Pick top M
top_M <- order(chi2_noisy, decreasing = TRUE)[1:M]

## Step 3: Re-noise the winners (fresh noise on true values)
chi2_private <- chi2_all[top_M] +
  rlaplace(M, scale = 2 * M / eps * sens_chi2)

## Post-process to Z-scores (free by DP composition)
Z_private <- sign(beta_hat[top_M]) *
  sqrt(pmax(0, chi2_private))
```

# DP in Action: Privacy vs. Statistical Power



**Key insight:** DP doesn't mean “no utility”. It means calibrated noise that preserves strong signals while protecting individuals.

# Recall: The eQTL Linking Attack

---

## The vulnerability:

- 1 Published expression data for anonymous samples
- 2 Public eQTL catalog (genotype  $\leftrightarrow$  expression)
- 3 Reference genotype database

## The attack:

Expression  $\xrightarrow{\text{eQTL}}$  Predicted genotype  $\xrightarrow{\text{match}}$  Individual ID

**Result:** With 50 genes, individuals can be uniquely identified

**Now:** Can differential privacy stop this attack?

# Can DP Protect Against eQTL Attacks?

---

**Setup:** Realistic scenario (50 genes, 500 individuals)

**Defense:** Add Laplace noise to expression values before publishing

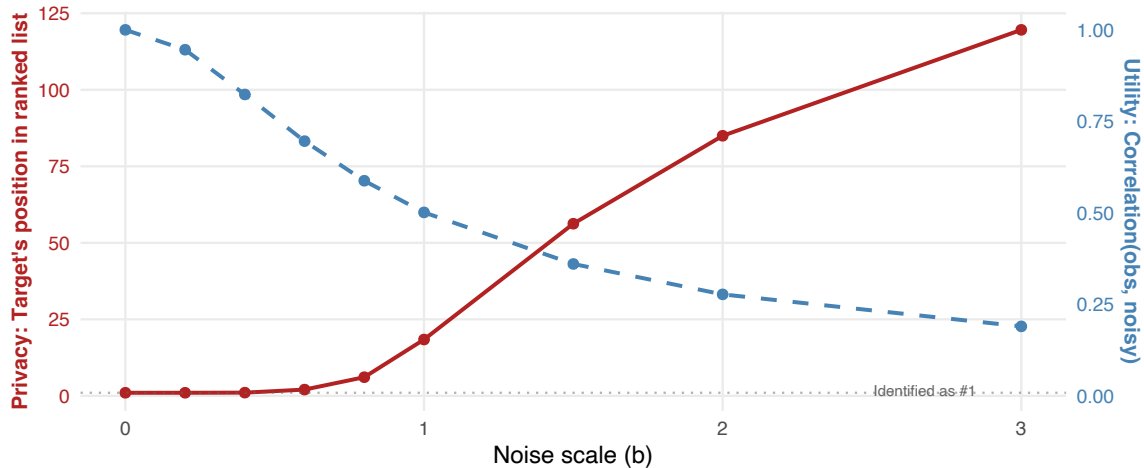
$$e_k^{\text{priv}} = e_k + \text{Lap}(b)$$

**Attack:** Predict genotypes from noisy expression, then match to database

**Question:** How much noise is needed to stop the attack?



# Does DP Stop the eQTL Attack?



# The DP Tradeoff for eQTL Data

---

## What we learned:

### Privacy side:

- No noise ( $b = 0$ ): Target at rank  $\sim 1$  (identified)
- Small noise ( $b < 0.6$ ): Rank  $\sim 5-10$  (still narrow)
- Moderate noise ( $b \geq 1$ ): Rank  $\sim 20+$  (lost in crowd)

### Utility side:

- No noise: Perfect correlation ( $r=1.0$ )
- Moderate noise: Correlation  $\sim 0.6-0.7$
- High noise: Data quality severely degraded

**Bottom line:** DP can protect individuals, but you pay with statistical power for discovery.

# Today's lecture

---

- 1 Genetic Data Utility vs. Privacy
- 2 Membership inference
- 3 Re-identification attack
- 4 Secured protocols for genomic data analysis
- 5 Differential Privacy
- 6 Summary**

# Comparison of approaches

|                    | DP                    | MPC            | HE            | FL       | TEE      |
|--------------------|-----------------------|----------------|---------------|----------|----------|
| Privacy guarantee  | Formal ( $\epsilon$ ) | Info-theoretic | Computational | Informal | Hardware |
| Utility loss       | Yes (noise)           | None           | None          | Some     | None     |
| Computational cost | Low                   | Moderate       | High          | Moderate | Low      |
| Multi-party native | –                     | Yes            | Limited       | Yes      | Limited  |

In practice these are **combined**: e.g., federated learning + DP noise + MPC aggregation.

# Genomic privacy

---

- ❶ **Summary statistics are not safe.** Membership inference from pooled allele frequencies.
- ❷ **Expression data leaks genotypes.** eQTL linking attacks on RNA-seq and scRNA-seq.
- ❸ **Defenses exist but involve tradeoffs.** No single mechanism suffices; layered approaches are the norm.

# Nevertheless, information has to be shared

---

- Why?
- How?
- Ownership?
- Should a government use it?

# References

---

- Erlich, Yaniv, Tal Shor, Itsik Pe'er, and Shai Carmi. 2018. "Identity Inference of Genomic Data Using Long-Range Familial Searches." *Science* 362 (6415): 690–94.
- Gürsoy, Gamze, Tianxiao Li, Susanna Liu, Eric Ni, Charlotte M Brannon, and Mark B Gerstein. 2022. "Functional Genomics Data: Privacy Risk Assessment and Technological Mitigation." *Nat. Rev. Genet.* 23 (4): 245–58.
- Gymrek, Melissa, Amy L McGuire, David Golan, Eran Halperin, and Yaniv Erlich. 2013. "Identifying Personal Genomes by Surname Inference." *Science* 339 (6117): 321–24.

# References

---

- Harmanci, Arif, and Mark Gerstein. 2016. “Quantification of Private Information Leakage from Phenotype-Genotype Data: Linking Attacks.” *Nat. Methods* 13 (3): 251–56.
- Homer, Nils, Szabolcs Szelinger, Margot Redman, et al. 2008. “Resolving Individuals Contributing Trace Amounts of DNA to Highly Complex Mixtures Using High-Density SNP Genotyping Microarrays.” *PLoS Genet.* 4 (8): e1000167.
- Sankararaman, Sriram, Guillaume Obozinski, Michael I Jordan, and Eran Halperin. 2009. “Genomic Privacy and Limits of Individual Detection in a Pool.” *Nat. Genet.* 41 (9): 965–67.



# References

---

- Shringarpure, Suyash S, and Carlos D Bustamante. 2015. “Privacy Risks from Genomic Data-Sharing Beacons.” *Am. J. Hum. Genet.* 97 (5): 631–46.
- Uhler, Caroline, Aleksandra B Slavkovic, and Stephen E Fienberg. 2012. “Privacy-Preserving Data Sharing for Genome-Wide Association Studies.” *arXiv [Stat.ME]*, May.
- Walker, Conor, Arjun Bhatt, Neda Ghaderi, et al. 2024. “Identifying Gene Expression Programs of Cell-Type Identity and Cellular Activity with Single-Cell RNA-Seq.” *Cell*.
- Yamamoto, Akito, and Tetsuo Shibuya. 2021. “More Practical Differentially Private Publication of Key Statistics in GWAS.” *Bioinform. Adv.* 1 (1): vbab004.